Cartesian products of graphs as spanning subgraphs of de Bruijn graphs* (Extended Abstract)

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Abstract. For Cartesian products $G = G_1 \times \ldots \times G_m$ $(m \geq 2)$ of nontrivial connected graphs G_i and the n-dimensional base B de Bruijn graph $D = D_B(n)$, we investigate whether or not there exists a spanning subgraph of D which is isomorphic to G. We show that G is never a spanning subgraph of D when n is greater than three or when n equals three and m is greater than two. For $n = 3$ and $m = 2$, we can show for wide classes of graphs that G cannot be a spanning subgraph of D. In particular, these non-existence results imply that $D_B(n)$ never contains a torus (i.e., the Cartesian product of $m \geq 2$ cycles) as a spanning subgraph when n is greater than two. For $n = 2$ the situation is quite different: we present a sufficient condition for a Cartesian product G to be a spanning subgraph of $D = D_B(2)$. As one of the corollaries we obtain that a torus $G = G_1 \times \ldots \times G_m$ is a spanning subgraph of $D = D_B(2)$ provided that $|G| = |D|$ and that the G_i are even cycles of length ≥ 4. In addition we apply our results to obtain embeddings of relatively small dilation of popular processor networks (as tori, meshes and hypercubes) into de Bruijn graphs of fixed small base.

Keywords: de Bruijn graphs, Cartesian product, graph embeddings, dilation, processor networks, parallel image processing and pattern recognition, massively parallel computers

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1 Introduction

In the context of parallel and distributed computation the problem of embedding one interconnection network into another is of fundamental importance and has gained considerable attention; see, e.g., the textbook of Leighton [3], the survey article of Monien and Sudborough [4], and the literature mentioned there. Nevertheless the field still offers various interesting open problems and sometimes even largely unexplored areas: for example, embeddings into de Bruijn networks have rarely been investigated until, recently, the topic was picked up by Heydemann et al [7] and, independently and with emphasis on applications in digital image processing, by two of the present authors [6]. Either of these papers discuss embeddings of certain Cartesian product networks into de Bruijn graphs: the paper of Heydemann et al [7] contains a variety of results on embeddings of hypercubes and 2–dimensional meshes into their optimal de Bruijn graphs, while meshes of higher dimension and tori are considered only briefly. In contrast, in the paper [6] the emphasis is on tori. (For definitions, see below.) For information on the role of de Bruijn networks in parallel image processing and pattern recognition, see e.g. [2].

In the present paper, for Cartesian products $G = G_1 \times \ldots \times G_m$ $(m \geq 2)$ of non-trivial connected graphs G_i and the n-dimensional base B de Bruijn graph $D = D_B(n)$, we investigate whether or not there exists a spanning subgraph of D which is isomorphic to G and apply our results to obtain embeddings of relatively small dilation of common processor networks (as tori, meshes and hypercubes) into de Bruijn graphs of small fixed base. Our results generalize and improve some of the results previously obtained in [6, 7].

The paper is organized as follows. In the remainder of this section, we collect some definitions and notational conventions. Thereafter, in Section 2, we prove some lemmas and in Section 3, we present our main results (Theorems 7 and 9). We close with a remark on embeddings into de Bruijn graphs of small fixed base and briefly report on experimental results.

Unless stated otherwise, graphs will be undirected, multiple edges and loops are admitted. Our terminology is standard and (essentially) in accordance with the terminology of [1]; graph-theoretic notions not defined here can be found in [1]. For a graph G, we denote by $V(G)$ and $E(G)$ the set of vertices and edges, respectively. The complete graph with n vertices is denoted K_n. The complete bipartite graph with color classes of cardinality n and m, respectively, is denoted $K(n, m)$. In directed graphs, we write $\overrightarrow{ab}$ to denote an arc from a to b. A *directed trail* (of length t) in a digraph $\overrightarrow{D}$ is a sequence $a_0, \ldots, a_t$ of vertices such that $\overrightarrow{a_i a_{i+1}}$ is an arc of $\overrightarrow{D}$ $(i = 0, \ldots, t-1)$ and such that all arcs $\overrightarrow{a_i a_{i+1}}$ are distinct. A graph is *non-trivial* if it has at least two vertices.

For graphs G_i $(i = 1, \ldots, m)$, the *Cartesian product* $G_1 \times \ldots \times G_m$ is the graph with vertex set $V(G_1) \times \ldots \times V(G_m)$ where two distinct vertices $(a_1, \ldots, a_m)$, $(b_1, \ldots, b_m)$ are joined by t (multiple) edges if and only if there exists an $i \in \{1, \ldots, m\}$ such that $a_j = b_j$ for all $j \neq i$ and such that there are t edges of G_i joining a_i with b_i; further, there are $\sum_{i=1}^{m} t_i$ loops at a vertex $(a_1, \ldots, a_m)$,

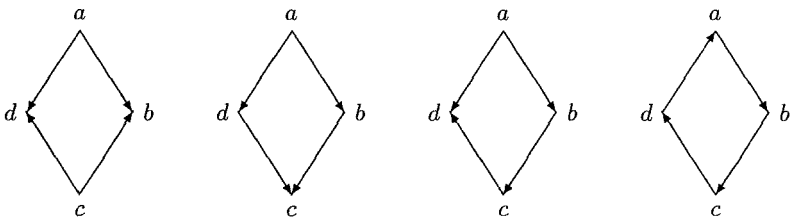


Fig. 1. The different types of 4-cycles in $\overrightarrow{D}_B(n)$.

where t_i is the number of loops at a_i in G_i.

Popular interconnection networks in the area of parallel computing are the tori, meshes and hypercubes which, in terms of the Cartesian product, can be defined as follows. A graph G is an *m-dimensional mesh* (*m-dimensional torus*) if G is the Cartesian product $G_1 \times \ldots \times G_m$ of m non-trivial paths (cycles). The *m-dimensional binary hypercube* is the graph $G = G_1 \times \ldots \times G_m$ where the G_i are complete graphs on two vertices ($i = 1, \ldots, m$). (For $m = 3$, G is called *cube.*)

For integers B, $n \geq 2$, the *directed base B de Bruijn graph of dimension n* (denoted $\overrightarrow{D}_B(n)$) has as its vertex set the set of n-tuples $a = \langle a_{n-1}, \ldots, a_0 \rangle$ where the entries a_i are integers with $0 \leq a_i \leq B-1$, and there exists an arc from $a = \langle a_{n-1}, \ldots, a_0 \rangle$ to $b = \langle b_{n-1}, \ldots, b_0 \rangle$ if and only if $b_i = a_{i-1}$ for $i = 1, \ldots, n-1$. The corresponding *undirected base B de Bruijn graph of dimension n* (denoted $D_B(n)$) results from $\overrightarrow{D}_B(n)$ by ignoring the orientation of the arcs. In other words, $D_B(n)$ has the same vertex set as $\overrightarrow{D}_B(n)$ and distinct vertices $a = \langle a_{n-1}, \ldots, a_0 \rangle$ and $b = \langle b_{n-1}, \ldots, b_0 \rangle$ are joined by a single edge if either $b_i = a_{i-1}$ for $i = 1, \ldots, n-1$ or $a_i = b_{i-1}$ for $i = 1, \ldots, n-1$, and by exactly two edges if both of these conditions hold; moreover, there is a loop at $a = \langle a_{n-1}, \ldots, a_0 \rangle$ if $a_0 = a_1 = \ldots = a_{n-1}$.

2 Some Lemmas on 4-cycles in de Bruijn graphs and Cartesian products

In this section we present a series of lemmas which prepare the proof of Theorem 7. The proof of Theorem 9 is independent of these lemmas. The reader who is mainly interested in results which are relevant for practical applications (Theorem 9 and its corollaries) may skip Section 2.

For a 4-cycle $C \subseteq D_B(n)$, let $\overrightarrow{C}$ be the corresponding directed subgraph of $\overrightarrow{D}_B(n)$. Then C (as well as $\overrightarrow{C}$) is called a *cycle of type t* if t is the maximum length of a directed trail contained in $\overrightarrow{C}$. Figure 1 displays the different types of 4-cycles.

In order to denote the vertices of a 4-cycle of $\overrightarrow{D}_B(n)$ and to indicate arc directions, we use notations like $a \to b \to c \to d \leftarrow a$ the meaning of which is clear without explanation.

Lemma 1. *$\overrightarrow{D}_B(n)$ contains no cycle of type 2.*

Proof. Suppose that $\overrightarrow{D}_B(n)$ contains a cycle of type 2, say, $a \to b \to c \leftarrow d \leftarrow a$ with $a = \langle a_{n-1}, \ldots, a_0\rangle$, $b = \langle b_{n-1}, \ldots, b_0\rangle$, $c = \langle c_{n-1}, \ldots, c_0\rangle$, $d = \langle d_{n-1}, \ldots, d_0\rangle$. Then $b_i = a_{i-1} = d_i$ $(i = 1, \ldots, n-1)$ and $b_0 = c_1 = d_0$, contradicting $b \neq d$. □

If $a \to b \to c \to d \leftarrow a$ is a cycle of type 3, then $a \to b \leftarrow c \to d \leftarrow a$ is called *the corresponding cycle of type 1*

Lemma 2. *If $\overrightarrow{D}_B(n)$ contains a cycle of type 3, then it also contains the corresponding cycle of type 1.*

Proof. Let $a \to b \to c \to d \leftarrow a$ be a cycle of type 3 contained in $\overrightarrow{D}_B(n)$ with $a = \langle a_{n-1}, \ldots, a_0\rangle$, $b = \langle b_{n-1}, \ldots, b_0\rangle$, $c = \langle c_{n-1}, \ldots, c_0\rangle$, $d = \langle d_{n-1}, \ldots, d_0\rangle$. Then $b_i = a_{i-1}$ $(i = 1, \ldots, n-1)$, $c_i = a_{i-2}$ $(i = 2, \ldots, n-1)$, $c_1 = b_0$, $d_i = a_{i-3}$ $(i = 3, \ldots, n-1)$, $d_2 = b_0$, $d_1 = c_0$, $d_i = a_{i-1}$ $(i = 1, \ldots, n-1)$. It follows that $b_i = d_i$ $(i = 1, \ldots, n-1)$ and consequently

$$\begin{aligned} b_i = d_i &= a_{i-3} = c_{i-1} \qquad (i = 3, \ldots, n-1) \\ b_2 = d_2 &= b_0 = c_1 \\ b_1 = d_1 &= c_0 \end{aligned}$$

which means that $\overrightarrow{D}_B(n)$ contains an arc from c to b. □

The next lemma is an immediate consequence of the definitions.

Lemma 3. *Let $a \to b \leftarrow c \to d \leftarrow a$ be a cycle of type 1 contained in $\overrightarrow{D}_B(n)$ with $a = \langle a_{n-1}, \ldots, a_0\rangle$, $b = \langle b_{n-1}, \ldots, b_0\rangle$, $c = \langle c_{n-1}, \ldots, c_0\rangle$, $d = \langle d_{n-1}, \ldots, d_0\rangle$. Then $a_i = c_i$ $(i = 0, \ldots, n-2)$ and $b_i = d_i (i = 1, \ldots, n-1)$.*

Lemma 4. *If $n \geq 4$ then every vertex of $\overrightarrow{D}_B(n)$ is contained in at most one cycle of type 4, and if $n = 3$, then every edge of $\overrightarrow{D}_B(n)$ is contained in at most one cycle of type 4.*

Proof. Let $a \to b \to c \to d \to a$ be a cycle of type 4 contained in $\overrightarrow{D}_B(n)$ with $a = \langle a_{n-1}, \ldots, a_0\rangle$, $b = \langle b_{n-1}, \ldots, b_0\rangle$, $c = \langle c_{n-1}, \ldots, c_0\rangle$, $d = \langle d_{n-1}, \ldots, d_0\rangle$. Put $a_{-1} := b_0$, $a_{-2} := c_0, a_{-3} := d_0$. Then $b_i = a_{i-1}$, $c_i = a_{i-2}$, $d_i = a_{i-3} (i = 0, \ldots, n-1)$ and $a_i = a_{i-4}$ $(i = 1, \ldots, n-1)$. In particular we have $a_{-3} = a_1$. Further, if $n \geq 3$ then $a_{-2} = a_2$ and if $n \geq 4$ then $a_{-1} = a_3$. Consequently, if $n \geq 4$ then the vertices b, c and d are uniquely determined by $a = \langle a_{n-1}, \ldots, a_0\rangle$. Further if $n = 3$ then c and d are uniquely determined by a and b. □

Lemma 5. *For $n \geq 4$ let $a = \langle a_{n-1}, \ldots, a_0 \rangle$, $c = \langle c_{n-1}, \ldots, c_0 \rangle$ be a pair of opposite vertices of a cycle of type 4 contained in $\overrightarrow{D}_B(n)$. Then, for at least one i with $1 \leq i \leq n-2$, we have $a_i \neq c_i$.*

Proof. Choose the notations as in the proof of Lemma 4 and, in addition, define $a_{-4} := a_0$. Then $a_i = a_{i-4}$ $(i = 0, \ldots, n-1)$. Suppose that $a_i = c_i$ for $i = 1, \ldots, n-2$. Then (because $c_i = a_{i-2}$ for $i = 0, \ldots, n-1$) we have $a_i = a_{i-2}$ $(i = 1, \ldots, n-2)$ and thus we conclude from $n \geq 4$ that $a_{n-3} = a_{n-5}$ and $a_2 = a_0$. Hence $a_{n-1} = a_{n-5} = a_{n-3} = c_{n-1}$, $a_0 = a_2 = a_{-2} = c_0$ and thus we have $a_i = c_i$ for all $i = 0, \ldots, n-1$ contradicting $a \neq c$. □

For a Cartesian product $G_1 \times \ldots \times G_m$, a subgraph H of $G_1 \times \ldots \times G_m$ is called *1-dimensional* if there exists an i such that, for any pair of vertices $a = (a_1, \ldots, a_m)$, $b = (b_1, \ldots, b_m)$ of H, $a_j = b_j$ for all $j \neq i$. A 4-cycle $C \subseteq G_1 \times \ldots \times G_m$ which is not 1-dimensional is called *2-dimensional.*

Lemma 6. *Let $G = G_1 \times \ldots \times G_m$ be the Cartesian product of $m \geq 2$ non-trivial connected graphs and let $\sim$ be an equivalence relation on the vertex set of G. Assume that $a \sim c$ for any two vertices a, c which form a pair of opposite vertices on some 2-dimensional 4-cycle of G. Then the partition of $V(G)$ corresponding to $\sim$ consists of at most two classes.*

Proof. Since every connected graph contains a spanning tree, it clearly suffices to prove the lemma for the case that the graphs G_i are trees. Then the graph G is bipartite since it is the Cartesian product of bipartite graphs. Let S be a color-class of G and pick $a, b \in S$. We claim that $a \sim b$ (which implies the lemma). Note that, in order to prove $a \sim b$, it suffices to consider the case when $d_G(a, b) = 2$ since the general case can be settled by iterated application of the distance-two case. Let $P = (a, x, b)$ be a path of G. If a, b is a pair of opposite vertices of a (2-dimensional) 4-cycle of G then we are done; otherwise P is a 1-dimensional subpath of G and (since the graphs G_i are non-trivial) there exists a 1-dimensional subpath $P' = (a', x', b')$ of G such that $P \cap P' = \emptyset$ and $aa', xx', bb' \in E(G)$. Hence $a \sim x' \sim b$. □

3 The main results

We now treat the question whether or not a Cartesian product of graphs is a spanning subgraph of $D_B(n)$. Our first theorem establishes a non-existence result for $n \geq 3$.

Theorem 7. *Let $G = G_1 \times \ldots \times G_m$ be the Cartesian product of $m \geq 2$ non-trivial connected graphs and, for some $n \geq 3$, let $D = D_B(n)$. If $m = 2$ and $n = 3$ then assume that G_1, G_2 are loopless, G_1 is not a path and G_2 has no vertices of degree one. Then G is not a spanning subgraph of D.*

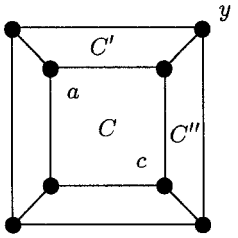


Fig. 2. Cube H contained in G in the case $m \geq 3$.

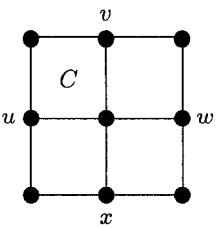


Fig. 3. 4-cycle C of type 4 in the case $m = 2$ and $n \geq 4$.

Let G_1 and G_2 be paths of length one and three, respectively. Then the mesh $G_1 \times G_2$ is a spanning subgraph of $D_2(3)$ which shows that the additional assumption made in Theorem 7 for the case $m = 2$, $n = 3$ cannot be dropped (but, possibly, relaxed). Due to space limitations, in this preliminary report we do not include the proof of Theorem 7 for the case $m = 2$, $n = 3$, the details of which are more involved; this part of the proof will be contained in the full paper.

Proof of Theorem 7 (under the additional hypothesis that $n \geq 4$ when $m = 2$). It suffices to settle the case that the graphs G_i are trees, which will be assumed henceforth. Suppose that G is a spanning subgraph of D.

First, let us assume that $m \geq 3$ or $|G_1|, |G_2| \geq 3$. We claim that the following holds.

$$\text{If } a = \langle a_{n-1}, \ldots, a_0 \rangle,\ c = \langle c_{n-1}, \ldots, c_0 \rangle \text{ is a pair of opposite vertices of a 4-cycle } C \subseteq G, \text{ then } a_i = c_i\ (i = 1, \ldots, n-2). \tag{1}$$

If C is not of type 4, then statement (1) is an immediate consequence of the Lemmas 1–3. Assume that C is of type 4. If $m \geq 3$ then C is contained in a cube $H \subseteq G$. Thus we can find 4-cycles C', C'' of H as shown in Figure 2. By Lemma 4 C' and C'' are not of type 4. Let $y = \langle y_{n-1}, \ldots, y_0 \rangle$ be the vertex of $C' \cap C''$ which is opposite of a in C' and opposite of c in C'' (cf. Figure 2). It follows that $a_i = y_i$ and $c_i = y_i$ $(i = 1, \ldots, n-2)$, which settles the case $m \geq 3$.

It remains to consider the case that $m = 2$. Then $|G_1|, |G_2| \geq 3$ and, therefore, C is contained in a mesh $H = P_1 \times P_2 \subseteq G$ where P_1 and P_2 are paths of length 2. Choose the notations as in Figure 3 and let $u = \langle u_{n-1}, \ldots, u_0 \rangle$, $v = \langle v_{n-1}, \ldots, v_0 \rangle$, $w = \langle w_{n-1}, \ldots, w_0 \rangle$, $x = \langle x_{n-1}, \ldots, x_0 \rangle$. Note that $n \geq 4$ by the additional hypothesis made at the beginning of the proof of Theorem 7. Thus we conclude from Lemma 4 that, with the exception of C, all 4-cycles of H are of type $\neq 4$. Hence (by the Lemmas 1 - 3) $v_i = w_i = x_i = u_i (i = 1, \ldots, n-2)$. On the other hand, by Lemma 5, we have $v_i \neq u_i$ for at least one $i \in \{1, \ldots, n-2\}$. This contradiction completes the proof of (1).

From (1) and Lemma 6 we conclude that the vertices of G can be partitioned into two disjoint classes such that $a_i = b_i (i = 1, \ldots, n-2)$ for any two vertices $a = \langle a_{n-1}, \ldots, a_0 \rangle$, $b = \langle b_{n-1}, \ldots, b_0 \rangle$ that are members of the same class. Hence

$$2B^2 \geq |G| = |D| = B^n$$

which implies $n = 3$, $B = 2$. Hence (by the assumption that $m \geq 3$ or $|G_1|, |G_2| \geq 3$) G is a cube, and thus we are done since $D_2(3)$ does not contain a cube as a spanning subgraph.

It remains to consider the case that $m = 2$ and, say, $|G_1| = 2$. Then G consists of two disjoint trees T_1, T_2 (which are isomorphic to G_2) together with edges e_v $(v \in V(T_1))$ where e_v connects v with a vertex of T_2 corresponding to v under a fixed isomorphism $T_1 \to T_2$. We refer to an edge e_v as a *rung* of G.

We first assume that $n \geq 5$. Let e be a rung of G connecting vertices $a = \langle a_{n-1}, \ldots, a_0 \rangle$ and $b = \langle b_{n-1}, \ldots, b_0 \rangle$ with $a_0 = \ldots = a_{n-2} = b_1 = \ldots = b_{n-1} = 0$; note that there exists at least one rung with these properties, namely, the rung containing the vertex $\langle 0, 0, \ldots, 0 \rangle$. Let C be a 4-cycle of G passing through e and let e' be the rung of C with $e' \neq e$. Denote by $a' = \langle a'_{n-1}, \ldots, a'_0 \rangle$ and $b' = \langle b'_{n-1}, \ldots, b'_0 \rangle$ the vertices incident with e' where notations are chosen such that a, a' is a pair of opposite vertices of C. Because $a_0 = \ldots = a_{n-2} = 0$ and $n \geq 5$, C cannot be a cycle of type 4; moreover one finds from $a_0 = \ldots = a_{n-2} = b_1 = \ldots = b_{n-1} = 0$ that $\overrightarrow{ba} \notin \overrightarrow{D} = \overrightarrow{D}_B(n)$. Hence $\overrightarrow{a'b}, \overrightarrow{a'b'}, \overrightarrow{ab'} \in \overrightarrow{D}$ by Lemma 1 and 2, and thus we conclude from Lemma 3 that $a'_0 = \ldots a'_{n-2} = b'_1 = \ldots = b'_{n-1} = 0$. Iterating this argument, we conclude from the connectedness of T_1 and T_2 that $v_1 = \ldots = v_{n-2} = 0$ for all vertices $v = \langle v_{n-1}, \ldots, v_0 \rangle$ of G, which is a contradiction. This settles the case $n \geq 5$.

Finally let $n = 4$. If a and b are vertices joined by a rung, then a and b are called *partners*. We claim that there exists a vertex $a = \langle a_3, a_2, a_1, a_0 \rangle$ with the following properties:

$$a_0 \neq a_2 \tag{2}$$

$$\overrightarrow{ab} \in \overrightarrow{D}, \text{ where } b \text{ is the partner of } a. \tag{3}$$

Indeed, pick a vertex $v = \langle v_3, v_2, v_1, v_0 \rangle$ with $v_1 \neq v_3$, $v_0 \neq v_2$ and let w be the partner of v. If $\overrightarrow{vw} \in \overrightarrow{D}$, then we are done; otherwise $\overrightarrow{wv} \in \overrightarrow{D}$, and thus $w = \langle v_4, v_3, v_2, v_1 \rangle$ for some $v_4 \in \{0, \ldots, B-1\}$ which (because $v_1 \neq v_3$) implies that w has the desired properties.

Now let $a =< a_3, a_2, a_1, a_0 >$ be a vertex with partner b such that (2) and (3) hold. Note that $\overrightarrow{ba} \notin \overrightarrow{D}$ since this, together with (3), would imply $a_0 = a_2$. Let $C = (a, b, c, d)$ be a 4-cycle contained in G. If C is of type 4, then $c = \langle a_1, a_0, a_3, a_2 \rangle$ and $\overrightarrow{cd} \in \overrightarrow{D}$, and thus c is (beside a) another vertex having the properties (2) and (3). If C is not of type 4, then one obtains from the Lemmas 1–3 (together with the fact that $\overrightarrow{ba} \notin \overrightarrow{D}$) that $c = \langle a'_3, a_2, a_1, a_0 \rangle$ and

$\overrightarrow{cd} \in \overrightarrow{D}$. Hence, in either case, c is a vertex having the properties (2) and (3). Iterating this argument, we conclude from the connectedness of T_1 and T_2 that every rung contains a vertex with the properties (2) and (3). On the other hand, this is clearly impossible since this cannot hold for the rung containing the vertex $\langle 0,0,0,0\rangle$. This contradiction settles the case $n = 4$. □

Note that Theorem 7 leaves open the case that $n = 3$ and G is a 2–dimensional mesh. (For results on embeddings of 2–dimensional meshes, we refer to [7].) For tori, however, Theorem 7 immediately implies the following.

Corollary 8. *A torus $G = G_1 \times \ldots \times G_m$ $(m \geq 2)$ cannot be a spanning subgraph of a de Bruijn graph $D_B(n)$ of dimension $n \geq 3$.*

For $n = 2$ the situation is quite different. In order to present our result for $n = 2$, we need some preparation. Let $G = G_1 \times \ldots \times G_m$ be a Cartesian product of $m \geq 2$ nontrivial graphs and $D = D_B(2)$. Let D and G have the same number of vertices, i.e.,

$$\prod_{i=1}^{m} |G_i| = B^2. \tag{4}$$

For each $i \in \{1, \ldots, m\}$, assume that $q_{i,1}, \ldots, q_{i,r_i} (r_i \geq 1)$ are integers with $q_{i,j} \geq 2$ $(j = 1, \ldots, r_i)$ such that $|G_i| = q_{i,1} \cdot \ldots \cdot q_{i,r_i}$. Assume further that the set $M = \{(i,j) : 1 \leq i \leq m, 1 \leq j \leq r_i\}$ can be partitioned into disjoint subsets M_1, M_2 such that, for every integer $q \geq 2, |\{(i,j) \in M_1 : q_{i,j} = q\}| = |\{(i,j) \in M_2 : q_{i,j} = q\}|$. Moreover, for each $(i,j) \in M$, let $a_{i,j}, b_{i,j}$ be positive integers with $q_{i,j} = a_{i,j} + b_{i,j}$, where $a_{i,j} = a_{k,l}, b_{i,j} = b_{k,l}$ whenever $q_{i,j} = q_{k,l}$. Put $t_{i,j} = (q_{i,j}, a_{i,j}, b_{i,j})$. We call a family of triples $t_{i,j} (1 \leq i \leq m, 1 \leq j \leq r_i)$ having the described properties a *family of triples associated to $G_1 \times \ldots \times G_m$*. Note that it follows from (4) that there exists at least one family of triples associated to $G_1 \times \ldots \times G_m$: just consider the factorization of $|G_i|$ into primes $q_{i,j}$ and choose the $a_{i,j}, b_{i,j}$ accordingly. One easily concludes from (4) that this results into a family of triples associated to $G_1 \times \ldots \times G_m$. Now the announced theorem reads as follows.

Theorem 9. *Let $G = G_1 \times \ldots \times G_m$ $(m \geq 2)$ be a Cartesian product of nontrivial graphs G_i and let $D = D_B(2)$. Assume that $|G| = |D|$ and let $t_{i,j} = (q_{i,j}, a_{i,j}, b_{i,j})$ $(1 \leq i \leq m, 1 \leq j \leq r_i)$ be a family of triples associated to $G_1 \ldots \times G_m$. Assume further that*

$$G_i \text{ is a spanning subgraph of } K(a_{i,1}, b_{i,1}) \times \ldots \times K(a_{i,r_i}, b_{i,r_i}), \quad i = 1, \ldots, m. \tag{5}$$

Then G is a spanning subgraph of D.

Proof. Let M, M_1, M_2 be as in the paragraph before Theorem 9. Then we conclude that

$$\bigtimes_{(i,j)\in M_1} K(a_{i,j}, b_{i,j}) \cong \bigtimes_{(i,j)\in M_2} K(a_{i,j}, b_{i,j}). \tag{6}$$

Let $H := \bigtimes_{(i,j)\in M_1} K(a_{i,j}, b_{i,j})$. Then it follows from (5) and (6) that G is a spanning subgraph of $H \times H$. Note that $H \times H$ is bipartite and connected; denote by C_0 and C_1 the color–classes of $H \times H$.

Let $\alpha : V(H \times H) \longrightarrow V(H \times H)$ be the function that maps (u, v) onto (v, u) for $u, v \in H$. Clearly α is an automorphism of $H \times H$. Since there exist vertices of $H \times H$ which are fixed by α, we conclude from the connectedness of $H \times H$ that α maps the vertices of C_i onto $C_i (i = 0, 1)$. Let $u, v \in V(H)$. We define a mapping $\Psi : V(H \times H) \longrightarrow V(D)$ by

$$\Psi((u, v)) = \begin{cases} \langle u, v\rangle & \text{if } (u, v) \in C_0 \\ \langle v, u\rangle & \text{otherwise .} \end{cases}$$

Because α maps C_1 onto C_1, we conclude that Ψ is a bijection. Let $(u, v), (x, y)$ be vertices of $H \times H$ which are joined by an edge. We may assume $(u, v) \in C_0, (x, y) \in C_1$. Then $\Psi((u, v)) = \langle u, v\rangle$ and $\Psi((x, y)) = \langle y, x\rangle$. Further $u = x$ or $v = y$. Hence $\langle u, v\rangle$ and $\langle y, x\rangle$ are joined by an edge of D and thus $H \times H$ is a spanning subgraph of D. Hence G is a spanning subgraph of D. □

We now present some corollaries of Theorem 9.

Corollary 10. *Let $G = G_1 \times \ldots \times G_m (m \geq 2)$ be a Cartesian product of even cycles of length ≥ 4 and $D = D_B(2)$. Assume that $|G| = |D|$. Then G is a spanning subgraph of D.*

Proof. Let $|G_i| = q_{i,1} \cdot \ldots \cdot q_{i,r_i}$ be the factorization of $|G_i|$ into primes and let $a_{i,j} = \lfloor \frac{q_{i,j}}{2} \rfloor$, $b_{i,j} = \lceil \frac{q_{i,j}}{2} \rceil$. Then (as mentioned above) $t_{i,j} = (q_{i,j}, a_{i,j}, b_{i,j})$ is a family of triples associated to $G_1 \times \ldots \times G_m$. Note also that each $K(a_{i,j}, b_{i,j})$ has a Hamiltonian path. Further, for each $i \in \{1, \ldots, m\}$ there exists a $j \in \{1, \ldots, r_i\}$ such that $K(a_{i,j}, b_{i,j}) \cong K_2$ (since $|G_i|$ is even). From this one easily concludes that the graph $K(a_{i,1}, b_{i,1}) \times \ldots \times K(a_{i,r_i}, b_{i,r_i})$ has a Hamiltonian cycle $(i = 1, \ldots, m)$, which means that (5) holds. Thus the assertion follows from Theorem 9. □

In a similar way one obtains

Corollary 11. *If $G = G_1 \times \ldots \times G_m (m \geq 2)$ is the Cartesian product of non-trivial paths G_i and if $|G| = |D_B(2)|$, then G is a spanning subgraph of $D_B(2)$.*

Note that the Corollaries 10 and 11 cover the cases of bipartite tori, arbitrary meshes and, as a special case of Corollary 11, binary hypercubes. The above results improve some of the results of Heydemann et al [7] who obtained the result of Corollary 10 under the additional assumption that the cycle lengths

are powers of two (see also [6]); similarly the paper [7] contains the result of Corollary 11 under additional assumptions on the path length.

further the authors of [7] present a restricted version of Corollary 11 covering the hypercube case. We mention that, in a sense, Corollary 10 is sharp since the Cartesian product $C_3 \times C_3$ is not a spanning subgraph of $D_3(2)$, as can easily be checked. (Here C_t denotes the cycle of length t.) In addition, we found that $C_5 \times C_5$ *is* a spanning subgraph of $D_5(2)$ but do not know whether or not $C_t \times C_t$ is a spanning subgraph of $D_t(2)$ for odd $t \geq 7$.

Other corollaries of Theorem 9 are possible. For example, the following can be derived from Theorem 9. (The proof is left for the reader.)

Corollary 12. *Let $G = G_1 \times \ldots \times G_m (m \geq 2)$ where the G_i are simple nontrivial bipartite graphs with color classes A_i, B_i and assume that $|A_i| = |A_j|, |B_i| = |B_j|$ whenever $|G_i| = |G_j|$. Assume further that $|G| = |D_B(2)|$ and that one of the following conditions (i) or (ii) holds:*

(i) $|G_i|$ is a prime for $i = 1, \ldots, m$

(ii) $|G_1| = \ldots = |G_m|$ and m is even.

Then G is a spanning subgraph of $D_B(2)$.

We close with a remark on embeddings into base C de Bruijn graphs where C is a fixed constant. An *embedding* of a graph G into a graph H is a pair of mappings $\{\phi_1, \phi_2\}$ where ϕ_1 maps each vertex of G onto a vertex of H and ϕ_2 assigns to every edge ab of G a path $\phi_2(ab)$ of H which connects $\phi_1(a)$ with $\phi_1(b)$. The *dilation* of an embedding is the maximum length of the paths $\phi_2(ab)$. If ϕ_1 is injective, then the embedding is said to have *load 1*.

Proposition 13. *For $n = 2\lceil \log_C B \rceil$ there exists an embedding $\{\phi_1, \phi_2\}$ of $D_B(2)$ into $D_C(n)$ with load 1 and dilation $\frac{n}{2}$.*

Proof. (cf. [5, 7]) Let $r = \lceil log_C B \rceil$. For a vertex $\langle a, b \rangle$ of $D_B(2)$, $0 \leq a, b < B$, let

$$\phi_1(\langle a, b \rangle) = \langle a_{r-1}, \ldots, a_0, b_{r-1}, \ldots, b_0 \rangle \in D_C(n)$$

with $a = \sum_{i=0}^{r-1} a_i C^i$ and $b = \sum_{i=0}^{r-1} b_i C^i$ where $a_i, b_i \in \{0, \ldots, C-1\}$ for $i = 0, \ldots, r-1$. Clearly ϕ_1 is injective. Then for any edge $\langle a, b \rangle \langle b, c \rangle \in E(D_B(2))$ there exists a path $(x_0, \ldots, x_r)$ in $D_C(n)$ with

$$\begin{aligned} x_0 &= \langle a_{r-1}, \ldots, a_0, b_{r-1}, \ldots, b_0 \rangle \\ x_i &= \langle a_{r-i-1}, \ldots, a_0, b_{r-1}, \ldots, b_0, c_{r-1}, \ldots, c_{r-i} \rangle \\ x_r &= \langle b_{r-1}, \ldots, b_0, c_{r-1}, \ldots, c_0 \rangle \end{aligned}$$

of length r from $\phi_1(\langle a, b \rangle)$ to $\phi_1(\langle b, c \rangle)$. Here $c = \sum_{i=0}^{r-1} c_i C^i$ with $c_i \in \{0, \ldots, C-1\}$. Thus the dilation of the embedding is $r = \frac{n}{2}$. □

In applications in the field of parallel algorithms, embeddings of relatively small dilation of common processor networks (as tori, meshes and hypercubes) into de Bruijn graphs of small fixed base (typically, 2, 3 or 4) are of significant

importance (cf. [2, 6]). Note that Theorem 9 (and its corollaries) in conjunction with the above proposition provides us with embeddings of this kind, where "relatively small dilation" means that the dilation is at most half the diameter n of $D_C(n)$. We refer to [6] for experimental results showing that this gain of a factor $\frac{1}{2}$ for the dilation leads to considerable gains in efficiency of parallel algorithms.

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